

Please attach this cover sheet to your completed homework

Homework 4 (Ch. 4)

ECE 311 – Hang Gao

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Problem	Course outcome	Grade
4.16	1.c.	/10
4.17	1.c.	/15
4.18	1.c.	/15
4.19 (a)	1.c.	/15
4.19 (b)	1.c.	/15
4.19 (c)	1.c.	/15
4.20	1.c.	/15
TOTAL:		/100

ABET Assessed Outcomes

1.c. Use appropriate models to formulate solutions related to electric circuit problems.

Homework #4

Due date: Sunday, March 1st by 11:59 PM (Electronically – Canvas)

- 4.16** A hydroelectric dam creates a reservoir of 10 km^3 . The average head of the reservoir is 100 m. Compute the PE of the reservoir.
- 4.17** A penstock is used to bring water from behind a dam into a turbine. The effective head is 20 m and the flow rate of water is $50 \text{ m}^3/\text{s}$. Compute the power of the water exiting the penstock.
- 4.18** The penstock of a hydroelectric power plant is 4 m in diameter. The water head is 60 m. Compute the water flow rate and the velocity of water inside the penstock if the power at the exit of the penstock is 10 MW.
- 4.19** A hydroelectric dam has a penstock that discharges 10^5 kg/s of water. The head of the dam is 80 m.
- Compute the volume of the discharged water per second (flow rate).
 - Compute the power of the water entering the turbine.
 - Assume that hydro efficiency is 0.95, the turbine efficiency is 0.9 and the generator efficiency is 92%, estimate the generated electrical power.
- 4.20** A natural gas power plant has a condenser that extracts 28,000 Btu/kg of natural gas. Compute the mechanical energy of the turbine and the overall system efficiency.